Measuring emergence through collapse of the effective joint possibility space

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Abstract

Emergence is claimed for scaled-model abilities, multi-agent coordination and animal collectives, yet each field certifies it differently, so identical behaviours can be labelled emergent or not. We define emergence as a spontaneous, persistent, regime-level collapse of the effective possibility space—the joint states, actions and trajectories still open to a collective. Collapse decomposes into environmental, individual, pairwise and higher-order sources. Calibrated on analytic ground truth and validated on off-design physics, the instrument shows that punctuated collapse accompanies learned conventions and roles whose value depends on collective adoption, whereas dense-reward controls without competing regimes collapse smoothly; that a capability can form smoothly across training while each deployment commits abruptly within an episode; and that remaining openness predicts intervention success prospectively. In a standard benchmark, perturbations alter conventions more often while the population is open than after commitment. On the same data, amplitude thresholds, information-theoretic composites and generic change-point detectors each fail at least one of these adjudications. Possibility-space collapse makes a contested concept measurable and falsifiable under a declared observation contract.

"More is different" describes interacting parts settling into a collective regime none specifies [1,2]. The word *emergence* names that phenomenon in statistical physics [3], in animal collectives [4–6], in the new capabilities of scaled neural networks [7], and in multi-agent reinforcement learning [8–10]. Yet the fields do not share an instrument: causal-emergence measures ask whether a coarse-graining has more causal power than its microstate [11–13]; autonomy measures test macro-variable self-determination [14]; partial-information decomposition asks how much of a target is carried synergistically [15,16]; phase-transition theory tracks an order parameter [17]; complexity science invokes irreducibility [18,19]; and machine learning reports "emergent abilities" as sharp benchmark upturns against scale [7,20], a debate entangled with scaling laws [21] and grokking [22]. Each family evaluates a chosen description—the causal power of a coarse-graining, the synergy of a target variable, the trajectory of a hand-picked order parameter. These approaches answer distinct questions; they do not jointly identify *when* a collective regime forms, *how abruptly*, and from *which* interaction structure.

What the fields share is not a measure but an intuition: an emergent phenomenon is unexpected beforehand yet lawful in retrospect—surprise has even been proposed as its defining test [23]. But surprise is an observer property, and a criterion that anything unfamiliar can satisfy names everything and measures nothing. Some reported emergent abilities are metric artefacts—change the score function and the sharp jump disappears [24]. The risk is defining emergence on the *observable of convenience*—a benchmark, an order parameter, a success rate—rather than on the object that reorganizes. Two systems can also produce byte-identical outcomes—identical joint distributions, marginals and task success—through entirely different couplings (Methods), and no statistic of that shared distribution can tell them apart. They become distinguishable only relative to a declared *observation contract*—which variables are measured, which are exogenous, which counterfactuals are admissible—under which one can ask how the effective possibility space closes.

The object is the joint space of what every part *could* still do together—the effective joint state–action–trajectory space. Before organization it is wide: an ant colony approaching a gap can still bridge, cross elsewhere or disperse[25]. Emergence, on our account, is the moment this space *collapses* onto a narrow committed regime—the colony decides, together, to build—and collapse is not completion: the bridge does not yet exist when the possibility space has contracted onto building it (Fig. 1). This definition keeps both halves of the surprise intuition and makes each measurable. In the punctuated examples studied here, low-order observables carry little warning during the open phase; after commitment begins, the collapse has reproducible timing, sources and consequences. Verdicts are stable within a measured admissible equivalence class of contracts (Methods).

Here we introduce a two-level definition of emergence: a qualification (spontaneous, regime-level, persistent) and a quantitative *emergence intensity profile* (EIP) whose axes—amplitude, abruptness and source decomposition—are measured separately, so abruptness is a phenotype, never an existence criterion. We then provide a calibrated measurement contract, matured through preregistered self-falsification, that recovers a collapse's source on analytic ground truth (72/72 cells) and off-design physics. In *learned* systems, collapse dissociates by timescale, varies systematically with system size and coupling, predicts controllability, and is **not** generic: the punctuated form arises spontaneously where learning must cross a joint exploration barrier, while ordinary dense-reward optimization collapses smoothly.

Results

Defining emergence through possibility-space collapse

We measure openness as the normalized entropy of the effective joint possibility distribution, $O_t = H(P_t)/H(P_{\text{ref}})$, and collapse as $C_t = 1 - O_t$. Because literal support never shrinks for a stochastic policy, we track the *effective* possibility count $N_{\text{eff}} = 2^H$, formalizing the intuition of a state space shrinking from 10^n to 2^n. A system *qualifies* as emergent when its collapse is regime-level (a structural, not marginal, change in the joint distribution), endogenous (internally generated, defined by a declared provenance boundary), and persistent (stable over the declared horizon). Robustness properties, such as recovery and causal irreversibility, are reported separately. For qualified systems, we report an intensity profile in place of a binary label. The distribution P_t is defined relative to a declared *observation contract*—variables, horizon, discretization, reference, and provenance boundary—that is frozen before adjudication. The contract makes the analyst's choices explicit; freezing, measured invariance and refusal to adjudicate outside its domain establish the tested range of validity (Methods).

Source decomposition then separates total collapse along a nested maximum-entropy ladder,

$$C_{\text{total}} = C_{\text{env}} + C_{\text{individual}} + C_{\text{pair}} + C_{\text{high}}, \quad (1)$$

making the interaction cut a *decomposer* rather than a gate, following connected-correlation expansions[26]. On analytic ground truth with four independent knobs, each knob moves only its own component. The ladder is monotone, and hiding an environmental common cause deterministically re-attributes collapse from C_{env} to the relational channels (five preregistered checks pass; Extended Data Fig. 1a). This makes contract-relativity precise: "higher-order" is defined only relative to what an observer declares exogenous. On a 72-cell factorial (4 sources $\times$ 3 temporal shapes $\times$ 2 stabilities $\times$ 3 magnitudes), the instrument recovers the correct source in 72/72 cells. Amplitude M is invariant across temporal shape (relative range 0.000) while abruptness J strictly orders punctuated $>$ sigmoid $>$ gradual (Extended Data Fig. 1b). This separation addresses the metric-artefact critique[24]; a revelation-only or discontinuous-metric control produces exactly zero collapse. A preregistered baseline battery then gives the standard information-theoretic toolkit its best a-priori shot on the same 72 cells: a composite rule built from marginal entropies, total correlation and pairwise mutual information recovers 54/72, misreading all 18 environment-driven cells as pairwise coupling, because a common cause and genuine coupling are indistinguishable to any functional of the system distribution without the contract's declared conditioning step. A raw entropy-amplitude threshold fails in both directions at once: it accepts the two external-takeover controls that the instrument rejects as exogenous, and accepts none of the 72 true positives (Methods; Supplementary Table 12).

Breakpoints mark phenotypes, not emergence itself

Qualification distinguishes emergent from non-emergent organization, while the *breakpoint* distinguishes the punctuated profile from gradual, non-monotone or unresolved collapse. A missing breakpoint therefore never rules emergence out by itself. We fit the collapse curve with a two-regime model against a one-regime fit using ΔBIC, inside a saturation-truncated window and behind an effect-size gate (two amendments forced by preregistered misses and fixed definitionally on fresh seeds). The matured detector was frozen and scored on a held-out labelled benchmark. It yielded zero false positives across deceleration, gradual and flat control families at every tested grid density and noise level, achieved an onset power of 1.00 at our operating resolution, and showed location agreement with a standard change-point method (Methods). Generic change-point detectors do not reproduce this operating point: calibrated to the same 5% false-positive budget on flat curves, binary segmentation and CUSUM both reach a power of 1.00 but fire on 100% of deceleration controls (binary segmentation also on 100% of gradual controls; Supplementary Table 12). The two-regime gate, not the change-point machinery, is what buys specificity. Every headline verdict below is additionally stable across 243 adjudication contracts (gate, window, truncation, threshold, grid) and across a preregistered battery of representations of the openness object itself. The measured equivalence class, including the representations that break it, is reported in Methods.

Using the matured detector, an *onset*-type breakpoint (slow $\rightarrow$ fast) follows the preregistered profile predictions: it is present where the theory predicts punctuated commitment and absent where it predicts smooth convergence. It appears in a committing ant colony, in a learned high-order coordination task (7/8 seeds across the original run and a five-seed extension), and at the Kuramoto synchronization transition[17,27]. It is absent in an ordinary supervised learner, whose collapse begins at maximum rate and decelerates (a *knee*, not an onset). It is unresolvable—and therefore not claimed—on stored language-model checkpoints, whose entropy collapses before the second saved step.

Formation and realization dissociate in learned systems

We next examine learned agents, whose possibility space is itself a product of optimization. In a two-phase collective-transport task, sixteen agents must jointly grip an object before they can push it, structurally delaying the choice of side. Policy-gradient training[28] learns this task in 10/10 seeds (success ≥ 0.995), all displaying the predicted signature: side-openness holds at a plateau for $\sim$17–19 steps and then collapses, with an onset breakpoint in every seed (ΔBIC 45.8–52.7, t^* at 16–18; Fig. 2b,d). A preregistered mechanism control—the same task without the gripping phase—shows no breakpoint (0/5). Thus, it is the structural delay, not learning per se, that produces punctuated collapse (Fig. 2c). The phenomenon reproduces under a different algorithm[29] (advantage actor–critic, byte-identical environment): 5/5 seeds reproduce the plateau-then-collapse shape and primary hinge (ΔBIC 37.7–45.5). This is a partial replication, as the strict thinning clause passes in only 3/5 seeds, missing solely due to thinned-subsample detector power.

This learned system separates two senses of emergence that are usually conflated (Fig. 3). Along the *realization* axis (within a single episode), collapse is punctuated (5/5 breakpoints). Along the *formation* axis (across training), the capability appears *smoothly*: outcome-openness expands ($0 \rightarrow \sim$0.65 within $\sim$100 updates) with no structural onset at either standard or fine 5-update resolution (0/5 on both). Formation is smooth and expansive, whereas realization is punctuated. This dissociation within a single system clarifies why debates over whether scaled-model abilities appear "suddenly" have been inconclusive[7,24]: a capability may form smoothly across training, while each deployment of it commits abruptly.

Punctuated collapse arises without designed barriers

The grip task imposes delayed commitment through its state machine and therefore serves as a positive control. We next preregistered two systems in which *no* environment mechanism delays commitment: every joint regime is available, and exactly equivalent, from the first update. In a ten-agent signalling population (five meanings, five symbols, reward only for successful decoding[30]), any of the 120 codes could be adopted at any moment. Yet convention openness holds at a statistically flat plateau and then collapses with an onset breakpoint in 4/5 seeds (ΔBIC 17.8–30.1). The breakpoint ($t^* = 275$–300) precedes mutual intelligibility 0.9 (updates 700–1,025) in every onset seed, with the five seeds converging to five *different* codes. A six-agent division-of-labour task (team reward only when all six interchangeable roles are covered) shows the same signature more strongly in 5/5 seeds (ΔBIC 53.6–71.7; closing slope $\sim 30\times$ the plateau slope), again with collapse preceding capability and five distinct role permutations.

In both systems the open plateau behaves as a joint exploration barrier, measured here through its unilateral component (Methods). Mid-plateau, a probe agent unilaterally committing to the *best* available code or role gains essentially nothing (0.011; 0.001). After commitment, the cheapest unilateral deviation forfeits the structural maximum of the payoff (0.40 of a 0.40 ceiling; 1.00), and populations from different seeds are mutually incompatible (cross-seed intelligibility 0.14, hybrid-team success 0.05, versus $\sim$1.00 within). A code or role division has value only insofar as others already share it. The collapse is nucleated, self-amplifying commitment—the learned analogue of sharp consensus transitions in naming games[31,32].

Neither system depends on its tabular parameterization. Randomly initialized neural policies (per-agent MLPs for signalling; one *shared* MLP for all six role agents) reproduce the signature: onset in 6/7 learned convention seeds and 10/10 role seeds (ΔBIC up to 162), collapse preceding capability in every case, and every seed converging to a different code or permutation. The neural runs expose a mechanism predicted by the theory: random initialization pre-breaks the symmetry among equivalent regimes, compressing the plateau roughly tenfold (Methods). An initialization-scale sweep moves commitment time in the predicted direction (monotone for signalling; role-system midpoint non-monotone, a registered miss). This mechanism—seed randomness deciding when and where a run commits—matches the finding that emergent language-model capabilities reflect bimodal cross-seed distributions rather than deterministic scale thresholds[33]. Each seed crosses the same barrier at a stochastic time, meaning ensemble averages blur what single-run collapse curves resolve. The convention and role results show that punctuated collapse does not require a designed gate or tabular parameterization. In these systems, barrier width depends on how symmetrically the competing regimes start.

Source typology transfers to learned coordination

Each channel of the ladder is realizable by learning (Fig. 4). A hidden-coordination task with eight individually-observing agents produces a purely *relational* collapse: per-agent marginals stay open (individual entropy $\approx$ 1.0 bit) while total correlation rises to 6.8/7 bits (Fig. 4a). When the low-order route is blocked with private information, learning builds a genuinely irreducible *higher-order* carrier: the textbook XOR structure emerges with C_{high} 0.94–0.96 bits and pairwise $\approx$ 0.0004 bits throughout the whole formation history (3/3 seeds; Fig. 4b). Declaring the private cues exogenous re-attributes the identical collapse to C_{env} (Fig. 4c), demonstrating the learned version of contract-relativity. In these tasks, higher-order structure is not unlearnable but *unfavoured*. Gradient learning selects the lowest-order implementation available and builds irreducible structure only when information constraints force it[34], echoing difficulties in emergent-communication[35–37] and cooperative-MARL[38,39] studies. Off design, the same ladder reads out Kuramoto oscillators (uncoupled $\rightarrow$ null; common driver $\rightarrow$ C_{env}; single edge $\rightarrow$ C_{pair}), a mechanism vocabulary it never saw. At matched task performance, the source *profile* separates a learned Overcooked pair[40] from a noise-matched scripted pair through its environment-conditioned component (C_{env} 0.0137 [0.0123, 0.0156] vs 0.0005 [0.0004, 0.0007]; Fig. 4d), even though a single-point coupling measure cannot. This difference is visible in the collapse composition, though not in any scalar summary.

Remaining openness predicts controllability

If emergence is the closing of a possibility space, then remaining openness should predict whether the outcome can still be steered. It does (Fig. 5). In the ant system, the per-episode flip-rate rises strictly with the episode's own remaining openness (0.000 $\rightarrow$ 0.205 across bins; closed episodes flip 0/2,600; flipped episodes are 0.58 openness units more open, permutation $p < 10^{-4}$ over 8,372 paired counterfactuals; Fig. 5d). In the learned grip system, a counter-regime impulse switches the outcome with probability 1.0 up to $t = 16$ and only 0.27 by $t = 30$, with side-openness predicting switchability with an AUC of 0.996 (Fig. 5a,b). This is not a time proxy. At a fixed intervention time, openness still separates flippable from locked episodes (AUC 0.974–0.990 at every tested τ; 20,480 episodes per cell). Conditioning additionally on the physical order parameter removes the signal, defining the registered boundary: here, openness is an information-sufficient transform of the physical state. Its value lies in being computable from the policy alone and transferring as a decision rule. It also works *prospectively*: a rule that waits until openness first drops to a threshold calibrated on the original seeds transfers to five unseen seeds, flipping 99.99% of episodes while acting 3.8 steps later than the best fixed schedule (99.6%) and beating random timing by 21 points. This provides a policy-accessible intervention coordinate, acting as late as each episode allows. A matched-parameter contrast isolates the mechanism. When stances are inertial (a hidden consolidation phase exists), joint openness predicts switchability *better* than the physical order parameter (AUC 0.886 vs 0.849). When the single stickiness parameter removes the hidden phase—with all other constants unchanged—the advantage reverses (0.811 vs 0.884; Fig. 5c). Both orderings were preregistered predictions. Openness carries controllability information beyond the order parameter exactly when there is a hidden regime to consolidate.

Abruptness scales with system size and criticality

The breakpoint is not specific to learning; it is a systematic feature of collective self-amplification. In an ant-commitment model, a single chooser *never* shows onset at any gain or grid, while large colonies always do, and the dependence is quantitative (Fig. 6a). Because commitment nucleates from binomial fluctuations of relative size $N^{-1/2}$ that are then amplified at an N-independent rate (a derivation recorded before running), commitment time should grow logarithmically while the collapse profile stays fixed. Both predictions hold across $N = 1$–500: $t_{50} = a + b \ln N$ with $b = 87.6$ (bootstrap 95% CI [41, 124], $R^2 = 0.93$). The closing width is N-invariant (280–290 trips from N=50 to 500), and the median collapse curves for $N \geq 50$ fall onto a single universal profile under pure time translation (mean pairwise RMS 0.010 versus 0.266 unaligned; Extended Data Fig. 2). This is sharper than equilibrium-style width rescaling[3]. One sub-prediction failed and is reported: at the longer horizon, the onset threshold sits between $N = 10$ and 20, so "onset at every $N \geq 10$" was counted as a miss. In the Kuramoto system, the same instrument tracks the approach to criticality. Across coupling $K = 0.9 \to 2.5$ with ten seeds per coupling (50/50 onsets), the mean breakpoint time falls strictly monotonically (6.64 $\to$ 1.80; Spearman $\rho = -0.99$, permutation $p = 10^{-5}$) and the mean closing slope rises strictly monotonically (0.030 $\to$ 0.197; $\rho = 0.98$, $p = 10^{-5}$), with per-coupling bootstrap intervals shown in Fig. 6b. Every subcritical run is gated null, and the collapse is carried by the pairwise channel (3/3 seeds; Fig. 6b)[41]. Abruptness therefore has two independent, manipulable control parameters—system size and feedback strength—both preregistered and confirmed.

Where the punctuated form does not appear

Onset-type collapse is **not** generic to learned optimization (Fig. 6c). In standard two-agent Overcooked, the learned policies collapse gradually on both a policy-entropy and a trajectory-occupancy object (0/3). A registered rescue sequence (smooth consensus, quorum populations, learning-rate scans) also produced no onset (0/20).

Cramped-room geometry pins the efficient plan, leaving no competing joint regimes. Introducing competing regimes *within the same standard package* restores commitment. In the official coordination-ring layout, clockwise and counterclockwise circulation are mirror-equivalent conventions with value only when shared. With identical PPO mechanics, all eight preregistered self-play seeds end committed to one direction (final direction probability 0.97 or 0.03; six counterclockwise, two clockwise), demonstrating endogenous symmetry breaking in an unmodified benchmark. In contrast, neither cramped-room control ever commits. The frozen detector certifies a punctuated onset in one seed (breakpoint at 780k steps preceding the capability crossing at 1M) and declines the rest. Their formation is non-monotone (commitment forms, dissolves, and re-forms), which the monotone-collapse instrument was not built to certify, so we count the onset-rate clause as a miss. The regime-level fact surviving every seed is the committed final convention.

A preregistered within-episode probe locates the commitment level: mid-training episodes drift with a direction bias but never internally lock (0/5), whereas final-checkpoint episodes are closed from the first step (initial openness 0.37, 8/8 seeds; untrained controls stay open, zero false onsets). The ring convention is a formation-level commitment with no realization stage, making it the mechanistic complement of the grip system, whose commitment lives within the episode.

A final preregistered experiment tests whether this commitment is causally real. We added unbiased Gaussian noise (two strengths) to checkpoints taken very early, at the most open point, and well after commitment. We then resumed training for 400k steps with byte-identical mechanics (8 seeds, 48 continuation runs; the seed is the independent unit). The preregistered primary endpoint—the perturbation flips or decommits the final convention—occurs in 7/8 seeds perturbed while open versus 1/8 after commitment (exact paired sign-flip $p = 0.016$; run-level 8/16 versus 1/16, Fisher $p = 0.008$). Strict direction reversal is secondary and descriptive (three open-phase flips, zero late; $p = 0.13$ across seeds). Stored openness at the perturbed checkpoint predicts strict flips (AUC 0.85; descriptive, runs cluster within seeds), and capability recovers to 0.92 of the unperturbed rate after late perturbation. What locks is the institution, not the learning. Because open and late checkpoints also differ in training time, we ran a preregistered fixed-time follow-up: all eight seeds perturbed at the same step (960k, selected by a variance-maximizing rule frozen in advance), where three seeds were still behaviourally open (openness 0.95–1.00) and five were committed. No run moved (0/16), a registered miss of the openness-predicts-movability clause. Every continuation, including the behaviourally open seeds, re-converged to its seed's eventual direction, and the open seeds' lower task maturity does not explain the null (Methods; Supplementary Table 13). Together the two experiments bound the claim. The checkpoint-conditioned contrast establishes a functional difference between empirically open and committed populations, but the common-clock control did not reproduce it, so openness is not isolated causally from the rest of training history. What the null does show is that the eventual direction is already parameter-committed while behaviour still looks open: formation-level commitment runs deeper than behavioural openness, and reading openness at a single fixed time is not a sufficient handle for redirection.

Among classic cases, the picture is mixed. Vicsek flocking[42], Swift–Hohenberg pattern selection[43], and Schelling segregation[44] organize strongly but collapse *gradually*, while the Potts model[45] reproduces the continuous-versus-first-order distinction through hinge strength and hysteresis. The result is a classification: collapse profiles sort systems into punctuated, gradual, parameter-axis, and early-warning forms. Onset appears specifically where a joint-regime barrier—physical (the grip gate), informational (the blocked low-order channel), or endogenous to learning (a convention or role with value only once shared)—must be crossed. Dense-reward optimization with no such barrier collapses gradually.

Discussion

Within a declared observation contract, we define emergence on the object that reorganizes—the effective joint possibility space. This choice avoids the failure mode behind the metric-artefact problem[24]. Amplitude and abruptness are measured separately and vary independently in the factorial calibration; a discontinuous scoring function produces zero collapse. The framework also supplies a source typology that transfers from analytic ground truth to physics to learned agents, a breakpoint whose presence and absence track the theory's predictions, manipulable dependences of abruptness on size and coupling, and a control-theoretic payoff: remaining openness predicts whether an intervention can still flip the outcome.

For machine intelligence specifically, the formation/realization dissociation clarifies a persistent confusion. "Emergent abilities" of scaled models[7] and the within-run appearance of a coordinated strategy in self-play agents[8,46,47] are different events on different timescales. Our learned flagship shows the first as smooth and expansive, and the second as punctuated. Controlled studies map when multi-agent language-model collaboration improves performance[48]; we instead ask when coordination becomes a persistent regime. Concurrent work detects higher-order structure in multi-agent language-model groups through partial information decomposition[49]. While that programme asks whether synergy is *present*, ours asks when the joint possibility space *closes*, how abruptly, through which channel, and whether the closure can still be steered.

Onset-type collapse is not generic to deep MARL at current scale: dense-reward Overcooked optimization collapses gradually. Our designed systems (the grip gate; the blocked-channel XOR task) are mechanism-isolating controls. Convention formation and role lock-in reproduce the signature with no designed barrier. What separates the two classes is a joint exploration barrier, whose unilateral component—no single-agent gradient onto the regime mid-plateau, a structural-maximum cost of leaving it after—is measured directly (Methods). The coordination-ring experiment tests this boundary inside the very benchmark that produced the negative: restoring competing regimes restores committed conventions in every seed. Certified onsets remain rare there because that commitment is non-monotone—an instrument boundary (Methods). Certifying non-monotone commitment and extending to large-scale self-play are the clear next steps.

Several boundaries remain. The commitment window is not causally verified: a one-pulse hypothesis failed its falsification clause, and formation proved re-entrant. Stored language-model checkpoints are too sparse for a breakpoint claim. Representation-independence holds only within a measured equivalence class and breaks where a representation erases the latent channel. Consistent with this boundary, lower-power or mis-declared regime objects usually lose verdicts, although one event-level convention object gains an onset where the declared object does not (Methods). With the analyst removed entirely, a frozen discovery recipe recovers the grip system's two-regime structure in every seed and its openness retains most of the fixed-time controllability signal (AUC 0.73–0.93 versus 0.98–0.99 declared), short of its registered bar: declaration buys measurement power, not the phenomenon. Nor is "emergence = onset breakpoint" a universal equivalence, as several canonical cases collapse gradually. The resulting instrument measures *how*, *when*, and *through which channel* a possibility space closes, while marking cases it cannot resolve. A natural next step is to connect the collapse object to internal-representation geometry in trained networks[50], extending the instrument from behaviour to mechanism.

Methods

Possibility space and openness. For a system of n parts we define the effective joint possibility distribution P_t over the relevant state–action–trajectory support at analysis time t, estimated from rollouts or exact enumeration where tractable. Openness is $O_t = H(P_t)/H(P_{\text{ref}})$ with H the Shannon entropy in bits and P_{ref} the maximum-entropy reference on the same support; collapse is $C_t = 1 - O_t$ and effective possibility count is $N_{\text{eff}} = 2^{H(P_t)}$. Entropies are estimated with a Miller–Madow-style bias correction and, for the joint objects, with the nested ladder below, which avoids direct high-dimensional estimation.

Intensity profile. For a collapse trajectory C_k, $k = 0, \ldots, T$ (entropy reduction from stage 0, in bits), with peak stage $k_{\text{p}} = \arg\max_k C_k$: amplitude is $M = C_{k_{\text{p}}}/H(P_{\text{ref}})$, the peak reduction as a fraction of the maximum-entropy reference; abruptness is $J = \max_k \delta_k / \sum_k \delta_k$ over the positive increments δ_k of C_k up to the peak—the largest single-stage share of the total collapse, so $J = 1$ for one-step collapse and $J \approx 1/T$ for uniform gradual collapse; the profile breakpoint estimate is the stage of the largest increment; and persistence requires $C_T \geq 0.5\, C_{k_{\text{p}}}$. These are the quantities calibrated in the 72-cell battery (Extended Data Fig. 1b). One convention keeps all units consistent: ladder components are raw entropy reductions in bits and telescope exactly, $C_{\text{env}} + C_{\text{individual}} + C_{\text{pair}} + C_{\text{high}} = H(P_{\text{ref}}) - H(P)$, where the ladder's stage-0 maximum-entropy distribution is exactly P_{ref}; the normalized quantities O_t, C_t and M divide by that same reference entropy.

Matched-confound battery. Four generators—a central script, a hidden common cause, pure coincidence and genuine local feedback—are constructed so that no statistic of the shared joint-action distribution separates them: joint distributions, single-agent marginals and macroscopic task outcomes are identical by construction. What separates them is the declared contract: conditioning on declared exogenous variables, interaction-severing counterfactuals and persistence checks assign each generator to its predicted certificate component. Attribution is therefore contract-relative, not mechanism discovery from outcome statistics.

Source decomposition. The nested maximum-entropy ladder constrains, in order: (i) the statistics of the declared exogenous variables and conditional independence given them (environment-conditioned); (ii) additionally each part's marginal (individual); (iii) additionally all pairwise marginals, following the connected-correlation construction of Schneidman et al.[26] (pairwise); and (iv) the full joint. C_{env}, $C_{\text{individual}}$, C_{pair} and C_{high} are the successive entropy reductions. Three properties matter for interpretation. First, each stage only adds constraints, so entropy is non-increasing along the ladder and every component is non-negative by construction. Second, the ladder is a sequence of information projections defined for *every* joint distribution—it is not a parametric model of the data—so "model misspecification" does not arise; a generator that mixes sources simply deposits entropy reduction at several levels at once, which is the intended reading. Third, the stage order is not a free knob: given a declared exogeneity boundary, the individual-to-pairwise-to-higher progression is the canonical hierarchy of marginal constraints, and the only genuinely movable choice—the boundary itself—is exactly the declared contract-relativity demonstrated in Fig. 4c. The provenance boundary is fixed before analysis in every experiment. A preregistered stress audit quantifies the remaining freedoms. Off-family generators the ladder was never built from are attributed correctly (a modular-sum triple with exactly independent pairwise marginals loads 3.32 bits on C_{high} with $C_{\text{pair}} = 0$; a Markov copy chain loads purely on C_{pair} with $C_{\text{high}} = 0$; fifty random Dirichlet joints give zero negative components and an exact telescoping identity). The constraint sets form a filtration, so no individual/environment reordering exists; the one genuine order freedom—declaring the environment before or after the pairwise stage—leaves the individual and higher-order components exactly invariant and moves the environment/pairwise split by at most 0.26 bits across an 81-cell grid. Finite-sample behaviour is measured, not assumed: median absolute component error is $\leq$0.018 bits at 3×10^4 samples for the three-agent test system, with no negative estimates in 120 runs, and the small-sample bias concentrates in C_{high} (0.31 bits at $n = 300$), which sets an explicit sample floor for higher-order claims. One registered audit prediction missed: when generator knobs interact strongly (pairwise copying under a parity mechanism), realized structure genuinely relocates across orders—copying makes the parity constraint first-order—so the ladder attributes it to the individual level. We report this as a property, not a defect: the ladder measures realized distributional structure, not generator labels.

Breakpoint detector. Collapse curves are fit with one- and two-regime piecewise-linear models; the breakpoint is accepted when the two-regime ΔBIC exceeds 10 and survives parity-thinning of the checkpoint grid, the curve passes an effect-size gate (max drop ≥ 0.1), and the analysis window is truncated at saturation (first point within 5% of the final value). An *onset* type requires $|\mathrm{slope}_{\mathrm{after}}| > |\mathrm{slope}_{\mathrm{before}}|$; the reverse ordering is a *deceleration* knee. The gate and saturation-truncation were added in response to two preregistered control failures (a flat subcritical false positive; a saturation-knee capture) and re-tested on fresh seeds.

Detector validation. Because the detector was matured through its own registered failures, it was frozen and then scored on an independent, preregistered synthetic benchmark of 200 curves per family (onset, deceleration knee, linear gradual, sigmoid, flat) per condition, with labels fixed before running. At the reference operating point (80 grid points, noise $\sigma = 0.02$) onset power is 1.00 and the false-positive rate across all control families is 0.000; power rises monotonically with grid density (0.00/0.62/0.995/1.00 at 12/20/40/80 points) while the false-positive rate remains 0.000 at every density, and degrades gracefully with noise (0.93 at $\sigma = 0.08$, false positives 0.000 throughout; Supplementary Table 1). Zero observed false positives among 200 curves per control family bounds the true rate at $\leq$0.5% per condition (95% rule-of-three over the pooled 600 controls); we report the bound rather than claiming the rate is exactly zero. The detector adjudicates monotone collapse only: a preregistered extension to non-monotone commitment (a settled-openness envelope recording irrevocable closure) kept false positives below 0.05 in every validation round but never exceeded 0.71 power against its required 0.90 on synthetic non-monotone positives, and under its own preregistered stopping rule was closed without ever being applied to system data; certifying non-monotone regime formation therefore remains an open instrument problem, which is why the coordination-ring onset count is reported at 1/8. A semi-synthetic injection audit quantifies the matching noise floor: commitments injected at known times into the ring pipeline's own estimator (real grids, real committed-episode counts, binomial direction noise) are recovered with false-positive rate 0.01 and median t^* error 1% of span, but detection power saturates at 0.88 (registered threshold 0.90, recorded as a near-miss)—the 30-episode evaluation budget itself caps single-curve power below the synthetic-library ceiling. Breakpoint locations agree with an independent standard change-point method (binary segmentation, RBF cost) to within 10% of span on onset curves; unlike generic change-point detection, the instrument correctly declines to call onset on deceleration and gradual curves. The zero power at 12 grid points sets the resolution floor: it is why no claim is made on sparse language-model checkpoint grids.

Method-baseline battery. A preregistered battery evaluates standard alternatives on exactly the data the instrument used, with all rival definitions frozen before implementation. On the 72-cell factorial, a composite of marginal entropies, total correlation and pairwise mutual information—given the fixed best-effort decision rule stated in the protocol—recovers the source in 54/72 cells and misassigns all 18 environment-driven cells as pairwise coupling; the composite has no environment rung because, without the contract's declared conditioning step, environmental and internal relational constraints are not separable by any function of the system joint distribution. A raw amplitude threshold ($C_{\mathrm{peak}} \geq 0.5$ of reference, no qualification) accepts the two external-takeover pseudo-controls the instrument rejects and accepts none of the 72 true positives. On the matched-confound mechanisms, joint entropy, total correlation and pairwise mutual information are equal across the three generators to machine precision while the contract verdicts differ. On the held-out curve benchmark, binary segmentation (RBF cost, gain acceptance) and CUSUM, each calibrated to a 5% false-positive rate on flat curves before scoring any other family, reach power 1.00 but fire on 100% of deceleration curves (binary segmentation also on 100% of gradual curves); neither attains the instrument's operating point (power 1.00 at false-positive rate 0.000). One preregistration detail was wrong and is retained: the protocol predicted the amplitude rule would accept the revelation/metric controls, but those have zero entropy amplitude and are rejected by the amplitude rule too; the clause passed through the external-takeover controls instead (Supplementary Table 12).

Adjudication-contract robustness. All headline breakpoint verdicts were re-adjudicated across 243 analysis contracts per system, crossing saturation fraction $\{0.02, 0.05, 0.1\}$, analysis window $\{0.75, 0.875, 1.0\}$ of the curve, effect-size gate $\{0.05, 0.1, 0.15\}$, ΔBIC threshold $\{8, 10, 12\}$ and checkpoint-grid stride $\{1, 2, 3\}$. For the grip flagship, the committing ant colony and the learned high-order task the onset verdict is unchanged in 100% of adequate-resolution cells and the breakpoint location varies by 1%, 8% and 11% of curve span respectively; aggressive grid coarsening reduces detectability (a power effect quantified by the validation benchmark above), never the location (Supplementary Table 2). The collapse object itself is fixed by the theory (the joint commitment space), not chosen by fit: the raw action-entropy of the grip system *rises* during the same window and is reported as such.

Observation contract. Every entropy is computed relative to a declared contract with six components: the variables admitted to the joint space, the trajectory horizon, the discretization, the reference distribution (maximum entropy on the same support), the probability-truncation rule, and the provenance boundary separating environment from system. Two properties anchor the contract's objectivity. First, an invariance that holds exactly: O_t is unchanged under any bijective recoding of states or actions, because Shannon entropy is permutation-invariant and the reference is defined on the same support—openness cannot be manufactured or destroyed by renaming. Second, an admissible class that is measured rather than assumed: representations remain admissible while they separate the competing regimes at or above the detector's resolution floor, and the equivalence battery below maps where that class ends (non-injective coarse-graining that merges regime-distinguishing states, or truncation that deletes the latent channel, breaks the verdict—mechanistically, not capriciously). The contract is frozen before adjudication and the instrument refuses outside it. One component deserves emphasis: the framework does not discover the regime variable from raw trajectories; it certifies a *declared* regime object. In every system here that object was named before analysis with a domain justification—side choice (bridge), grip configuration and push side (transport), code mapping (signalling), role permutation (division of labour), XOR carrier (blocked channel), circulation direction (Overcooked ring), phase locking (Kuramoto), magnetization sector (Potts)—and where no credible regime object exists the correct output is refusal, not a verdict. The instrument is thus a general measurement contract for declared collective regime objects, not an automatic emergence detector for arbitrary raw systems. The analyst's freedom is thereby the same freedom every measurement has—the choice of ensemble in thermodynamics, of coordinates in mechanics—made explicit, fixed in advance, and accompanied by its measured range of validity.

Representation equivalence. Detector settings and measurement representations are different freedoms, so they were audited separately. On byte-identical retrained seeds we recomputed the openness object under twelve preregistered representations—for the convention system, the population-mean mapping, per-agent mean entropy, the listener-side dual, a 2,048-sample behavioural estimate, a checkpoint-zero empirical reference, $\varepsilon = 0.01$ truncation and 5$\rightarrow$3 symbol coarse-graining; for the grip system, policy probabilities, state quantization at 0.1 and at 0.5, realized 16-agent action counts, and $\varepsilon = 0.01$ truncation—and adjudicated each with the frozen detector. The onset verdict survives in 8/12 cells and the breakpoint location is essentially representation-invariant wherever a verdict exists: $t^* = 250$–300 of a 4,000-update span (1.25%) in the convention system, and identical ($t^* = 17$) in every preserving grip representation. The registered ≥90% preservation bar was missed, so per the registered clause we report the measured equivalence class instead of claiming blanket invariance. All four breaking cells fail for identifiable reasons and still locate the commitment at $t \approx 19$–22: two fall just below the conservative ΔBIC$= 10$ bar while agreeing on location (per-agent entropy, an individual-level object distinct from the joint convention; 5→3 binning); 0.5-level state quantization compresses the transition into a one-to-two-checkpoint cliff that fails the frozen thinning clause (a resolution limit, not a disappearance: the unthinned fit gives ΔBIC 111, onset-type); and ε-truncation zeroes the side channel while it is latent (<1% of action mass during the grip phase), making the object non-monotone—truncating small probabilities destroys precisely the still-open possibilities the object exists to measure. The equivalence class is therefore: verdict and location survive object dualities, behavioural sampling, reference changes and state coarse-graining down to the detector's resolution floor; they break only when the representation erases the latent channel or compresses the transition below that floor (Supplementary Table 3).

Regime-object audits. The declared regime object is an analyst choice, so two further preregistered audits test whether verdicts depend on it. An *ensemble* audit enumerates, from each environment specification alone, the alternative regime variables an independent analyst could plausibly have declared (convention: the listener code, the composed speaker–listener channel, a single-meaning sub-regime; roles: one agent's role, one role's owner; grip: the realized force direction), together with control variables that erase the symmetric competition or are exogenous by construction, and adjudicates every candidate with the frozen detector on byte-identical reruns. On the formation axis, admissible alternatives reproduce the declared verdict in 22/25 cells (roles 10/10, listener code 5/5) and every co-certified breakpoint falls within the detector's span tolerance (19/19); the disagreements are one sub-regime cell below the evidence threshold and the composed channel in two seeds—a two-stage regime whose late second collapse the monotone detector declines. None of the 20 control cells certifies an onset. A *discovery* audit removes the analyst: one fixed recipe (k-means on raw episode records, k by silhouette, no per-system tuning) defines regime variables from behaviour alone. On the formation axis it reproduces 6/10 verdicts, with both convention disagreements at the detector threshold (ΔBIC 6.9 and 9.8 against 10) and one roles cell failing only the thinning clause; on the realization axis, state-trace and windowed-force proxies qualify the collapse but miss the strict onset bar in all three registered constructions (best ΔBIC 12.6), and untrained grip controls certify onset because the structural gate delays commitment for any policy—the mechanism control's own prediction. The registered agreement targets were missed (ensemble 22/30 against 24/30; discovery 6/15 against 12/15) and are reported with this diagnosis. Across all 105 audit cells the failures are one-directional: mis-declared or lower-power regime objects lose verdicts, and exactly one gains one (the event-level convention variable in the seed whose declared object shows no onset, ΔBIC 23.6). The declaration is a power parameter, not a cherry-picking lever (Supplementary Tables 4 and 5). A functional follow-up asks whether the machine-discovered regime variable still carries the controllability information with the analyst removed entirely: racing its openness against the declared side-openness on the same 81,920 grip intervention episodes (frozen protocol, appended preregistration), the discovered predictor reaches fixed-time AUCs of 0.73–0.93 against 0.98–0.99 for the declared object, missing the registered 0.80-everywhere bar (a registered miss), while raw policy entropy carries almost none of the signal (pooled AUC 0.62 against 0.996). The discovery recipe recovers the two-regime structure in 5/5 seeds; what declaration buys is measurement power, not the existence of the signal (Supplementary Table 14).

Emergence certificate. The frozen instruments are packaged into a single standardized adjudication script (`emergence_certificate.py`) so that a new system receives a reproducible verdict: a three-condition qualification (regime-level: the collapse gate *and*, where the source analysis is available, joint-beyond-marginal reorganization; endogenous: the declared provenance boundary; persistent: no re-opening beyond 0.1 of reference within the observed window), the intensity profile as a vector (amplitude, abruptness class, sharpness, t^*, source ranking), and a categorical verdict with an explicit *unresolvable* class below the measured power floor and a low-power flag between 12 and 40 grid points. We deliberately do not output a scalar "emergence score": amplitude and abruptness are demonstrated independent axes, and any fixed weighting of them would reintroduce the observable-of-convenience failure this framework exists to remove. Applied to the paper's systems, the certificate issues "emergent: punctuated" for the grip, convention, role and $N{=}100$ ant systems; the single-ant control—with nearly the same amplitude (0.92) as the colony—*fails* qualification, because its collapse is carried entirely by the individual source, with no joint regime involved, and the Overcooked occupancy object falls below the collapse gate with a low-power flag. Same entropy drop, different verdicts: the discrimination is exactly what a bare entropy measure cannot provide (per-system qualification table in Supplementary Table 6).

Learned systems. The grip-transport flagship uses 16 agents, grip threshold 6, a two-phase grip-then-push dynamics (grip gain 0.06, decay 0.01), trained by REINFORCE[28] (1,200 updates, batch 512, learning rate 2×10^{-3}) and, for the robustness check, by advantage actor–critic[29] on identical environment parameters (per-seed statistics in Supplementary Table 7). The hidden-coordination flagship uses 8 individually-observing agents with per-agent stances; the sticky variant sets stance-change probability 0.25 and the matched control sets it to 1.0 with all other constants imported unchanged. Overcooked runs use the standard cramped-room layout[40]. The non-constructed convention system is a ten-agent Lewis signalling population[30] (five meanings, five symbols, tabular policies, REINFORCE, 512 random speaker–listener pairs per update, 4,000 updates); the role system is six agents choosing among six roles with team reward only for full coverage (6,000 updates, otherwise identical training); in both, openness is computed from policy probabilities at 25-update checkpoints and adjudicated by the frozen detector (Supplementary Table 8). The neural replications keep both environments and recipes fixed and replace the tables with one-hidden-layer networks (32 tanh units, Adam 3×10^{-3}, ten fresh seeds per system): per-agent speaker and listener MLPs for the signalling population, and a single MLP shared by all six role agents, so that role differentiation must emerge inside one network. Because random initialization compresses commitment into the first few hundred updates, the neural runs are evaluated at 5-update resolution—a preregistered measurement-density amendment justified by the detector's validated grid-power curve; at the original 25-update grid the onsets are under-resolved (1/7 and 1/10), which is recorded alongside the fine-grid result (Supplementary Table 9). The initialization sweep varies the input-weight scale $\sigma \in \{0.02, 0.1, 0.45\}$ at five seeds per cell. The coordination-ring study trains standard `overcooked_ai` self-play PPO (mechanics and budget identical to the cramped-room runs, 2M steps) on the official `coordination_ring` layout, eight seeds, checkpoints every 20k steps; the regime object is the entropy of the pair's circulation direction (net winding around the central counter, Laplace-smoothed over 30 evaluation episodes per checkpoint), declared before running (per-seed conventions in Supplementary Table 10). A parallel preregistered attempt on the unmodified MPE `simple_spread` benchmark did not reach its frozen competence precondition (conflict-episode coverage 3% after 3,000 PPO updates) and is recorded as a training failure with no theory claim made either way. A planned second-layout study on the official `counter_circuit` layout, whose central circuit supports the same two mirror-equivalent circulation conventions, was stopped at the same gate: in three disclosed one-seed pilots (4M steps with the standard shaping anneal; 8M with a lengthened anneal; 6M with shaping never annealed) the fixed training recipe delivers zero soups, so no confirmatory protocol was frozen. In the annealed pilots a shaped-reward circulation habit forms and then dissolves once shaping expires—a convention without realized task value, which is precisely what the competence precondition exists to exclude. The causal commitment test perturbs stored ring checkpoints at 100k steps, at the seed's most-open checkpoint (last with direction probability in $[0.3, 0.7]$ and ≥ 20 committed evaluation episodes), and at 1.6M steps, adding per-tensor Gaussian noise of 0.25 or 0.5 relative standard deviations, resuming training for exactly 400k steps with unchanged mechanics and schedule position, and scoring the final convention against the seed's unperturbed direction; all times, scales, margins and clauses were frozen in the protocol file before any run. Because the two noise scales share a training seed, continuation runs cluster within seeds; the run-level Fisher test was the frozen clause, and a stricter seed-level paired analysis (exact sign-flip permutation over the eight seeds, labeled post-hoc in the protocol file) confirms the primary endpoint and demotes the strict direction-reversal contrast to descriptive. Because open and late checkpoints differ in training time as well as openness, a preregistered fixed-time follow-up perturbed all eight seeds at the same step, 960k, chosen by a frozen rule (the common-grid checkpoint maximizing cross-seed openness variance with ≥ 20 committed episodes per seed), with identical noise scales, resume length and scoring. No continuation moved (0/16; seed-level Fisher $p = 1.0$), a registered miss of the openness-predicts-movability clause at fixed time. The open seeds' lower checkpoint task performance (0.27–0.53 versus 0.67–1.9 mean soups) rules out maturity as an explanation of the null, and every continuation re-converged to its seed's eventual direction. The finite-size study uses sizes $N \in \{1, 2, 5, 10, 20, 50, 100, 200, 500\}$, 30 episodes per size and a 1,500-trip horizon, with t_{50} and width read from median openness crossings and the log-law slope bootstrapped over episodes (Supplementary Table 11). The ant, Kuramoto, Potts, Vicsek, Swift–Hohenberg and Schelling models follow their standard definitions (Supplementary Methods).

Barrier measurement. The joint-exploration-barrier claim is tested by deterministic replay of both learned positive systems with policy snapshots at every checkpoint (replay validity: all ten re-run seeds reproduce the stored final codes and role assignments exactly). At the checkpoint nearest $t^*/2$ (onset seeds only), the unilateral adoption gain of agent i is the payoff of a probe hard-committed to the best of all $K!$ codes (respectively all R role locks) against the unmodified population, minus the payoff of i's own current policy, averaged over agents; payoff is symmetric expected intelligibility with a random partner (convention) or the permanent of the row-stochastic role matrix (roles). At the final checkpoint the unilateral deviation cost is the payoff on the population's majority regime minus the best payoff over all alternative committed regimes. Cross-play pairs speaker populations with listener populations from different seeds (20 ordered pairs) and assembles hybrid role teams from every proper subset split of every distinct-assignment seed pair (620 teams). All four outcome clauses were frozen before the run; one threshold-calibration miss (the convention deviation-cost clause set 0.50 where the payoff structure caps the cost at $0.4\times$within-population payoff for $K{=}5$ permutation codes, the measured value 0.3996 sitting at that ceiling) is reported verbatim in the protocol file. What these quantities measure is a *unilateral coordination barrier*—the absence of single-agent payoff gradient toward any regime, and the single-agent cost of leaving one—not the full performance landscape over joint policy paths; multi-agent coordinated escape routes are not enumerated.

Scalability. The pipeline's cost is dominated by openness estimation, not by the detector (which fits two linear models per gridpoint and is linear in grid length). Openness requires the entropy of the declared possibility object, estimated either exactly from policy probabilities (learned tabular and neural systems), or from empirical distributions over the declared coarse-graining (physical systems), whose sample complexity we measured directly: in the finite-sample audit the median absolute error of every ladder component is $\leq$0.02 bits at 3×10^4 samples, rising at 300 samples to 0.09 bits (individual), 0.31 (higher-order) and 0.81 (pairwise), so the pairwise term is the first casualty of scarce data. Total openness itself is linear in population size times horizon, and the breakpoint stage ran unchanged at population sizes up to $N{=}500$ in the finite-size study; the source decomposition is the only superlinear stage (pairwise marginals cost $O(N^2)$ and the higher-order maximum-entropy fit is the practical ceiling), so qualification and profile can always be computed at scales where the full typology cannot. All experiments in this paper run on CPU except the Overcooked PPO training; no single run exceeds a day of wall-clock time.

Preregistration and integrity. Every predicted outcome was frozen, with an explicit falsification clause, in an internal protocol file before the corresponding run; passes and misses are reported identically, and each amendment is recorded in place with its trigger (the complete ledger is collected in Supplementary Note 1). We characterize these as internally time-stamped frozen protocols rather than third-party registrations: they were not lodged with an external registry, and we release them verbatim—including every miss, every failed instrument extension and every post-hoc analysis labeled as such—together with the code, raw outputs and an output manifest, so that the ordering of protocol and result can be audited from the released record. All numbers in the main text and figures are produced by a single extraction script run directly against the raw output files; no value is transcribed by hand.

One float32 numerical defect (an entropy clamp that rounded to a degenerate value) was found, fixed in code, and both affected experiments were rerun with identical seeds, with the corrupted outputs archived. No thresholds were changed after seeing results.

Reporting summary. Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article. The use of large-language-model agents is disclosed in the dedicated section in the back matter.

Data availability

The raw output of every experiment (one JSON file per registered run), together with the frozen protocol files that fix each prediction and falsification clause before running, is provided in the accompanying repository and will be archived with a permanent DOI on Zenodo upon acceptance. Source data for each main-text figure are generated directly from these files by the plotting script; no intermediate hand-edited data are used.

Code availability

All simulation, training, analysis and figure-generation code is available in the accompanying repository and will be archived on Zenodo upon acceptance. Every number, figure panel and Supplementary Table in the manuscript is regenerated from the raw stored outputs by scripts documented in the repository.

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Author contributions

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Competing interests

The authors declare no competing interests.

Use of large language models

Large language models were used, under the authors' direction and continuous review, to assist in writing and debugging code and in drafting and editing text. The experimental designs, protocols and adjudication criteria were specified by the authors, who reviewed and validated all code, protocols and analyses and take full responsibility for all claims.

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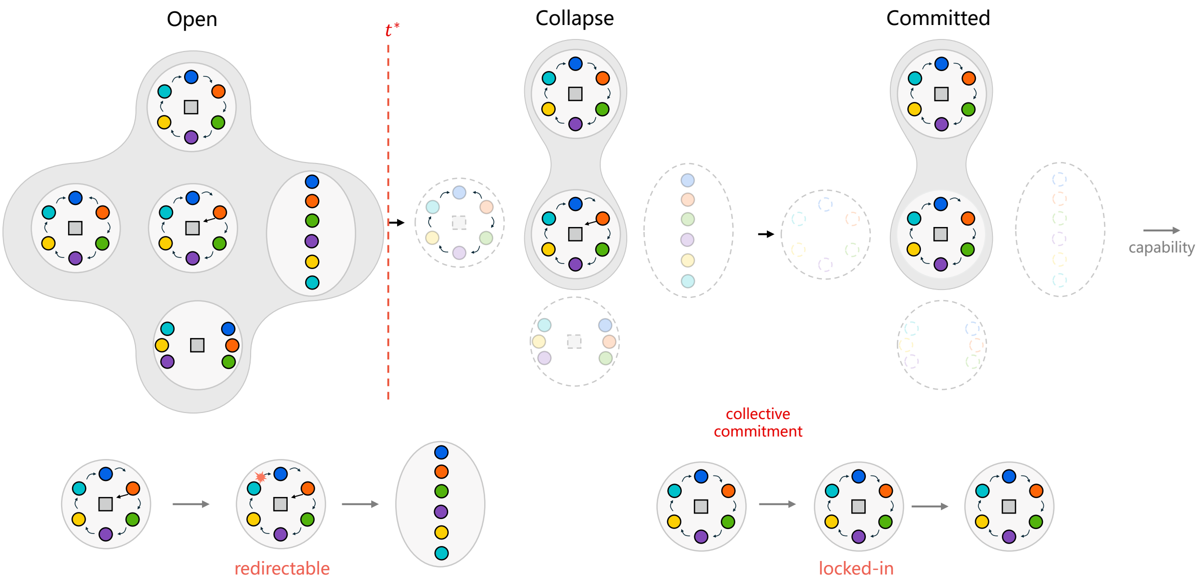


Figure 1. Emergence as possibility collapse. A population of learning agents (coloured dots) starts with many joint futures open (grey): several collective regimes—here a ring and a line—are still reachable. At a breakpoint t^* the joint possibility space begins to contract; alternatives fade until the population is committed to a single regime. In the learned convention and role systems, the measured capability criterion is crossed only after this commitment. Bottom, the practical consequence: while the space is open, a perturbation (red arrow) can redirect the population to a different regime; after commitment it is substantially less likely to do so.

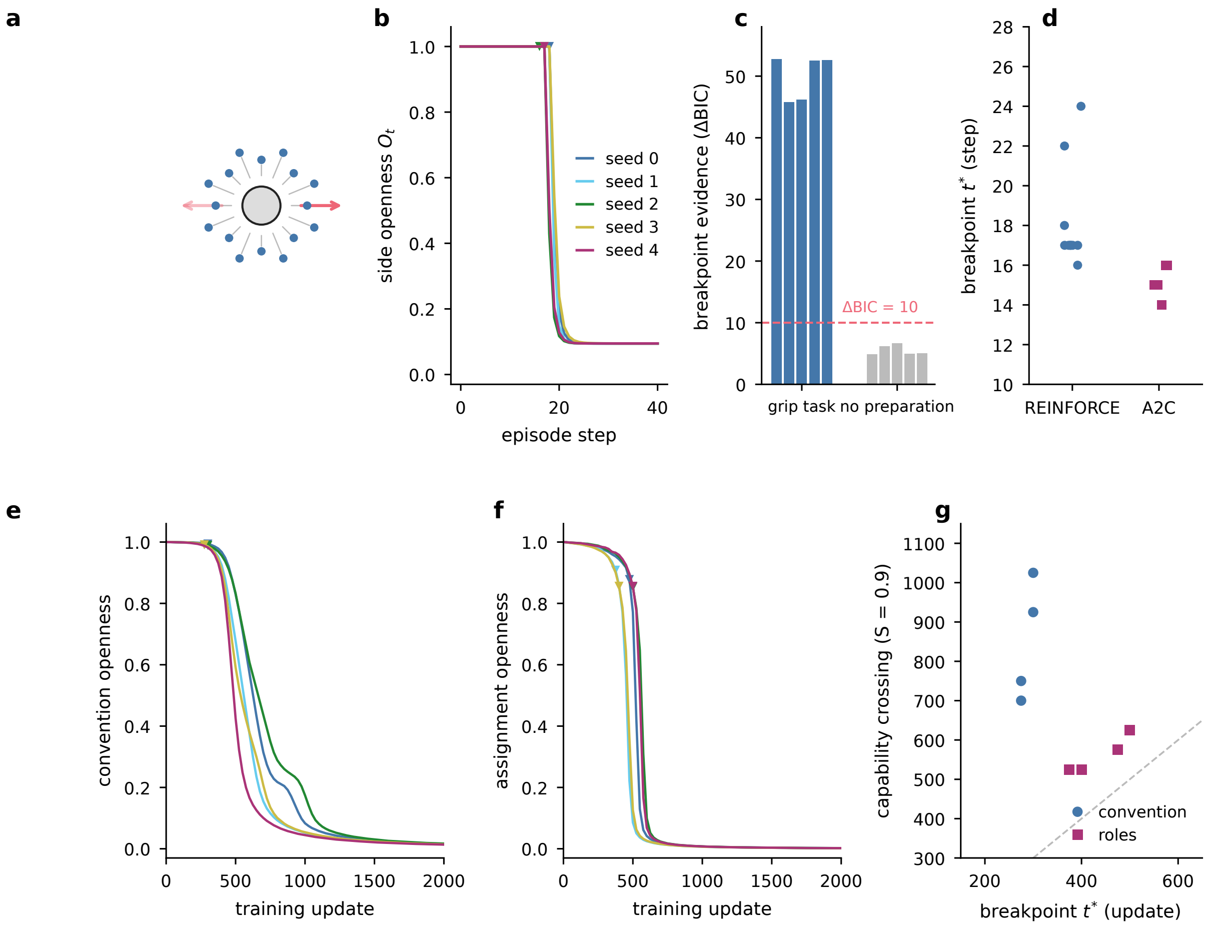


Figure 2. Punctuated commitment in learned collectives. **a**, The grip-then-push transport task ($N = 16$): agents must jointly grip before the object can be pushed (arrows, the two possible push directions), structurally delaying the choice of side. **b**, Side-openness of the joint policy holds on a plateau and then collapses; triangles mark the fitted breakpoint t^* (5 REINFORCE seeds). **c**, Mechanism contrast: the grip task shows a breakpoint in 5/5 seeds (ΔBIC well above 10) whereas a matched no-preparation task shows none (0/5). **d**, Breakpoint location across seeds and algorithms: 10/10 for REINFORCE, and a partial replication under A2C (5/5 shape and primary hinge, 3/5 strict). **e**, A ten-agent signalling population with 120 exactly equivalent codes and no gate holds a flat open plateau, then commits with an onset breakpoint (4/5 seeds; triangles), each seed to a different code. **f**, Six agents locking into a division of labour among 720 equivalent role permutations (5/5 onset, five distinct permutations). **g**, In all nine onset seeds the possibility-space breakpoint precedes the capability crossing ($S = 0.9$): commitment is measurable before the ability is expressed.

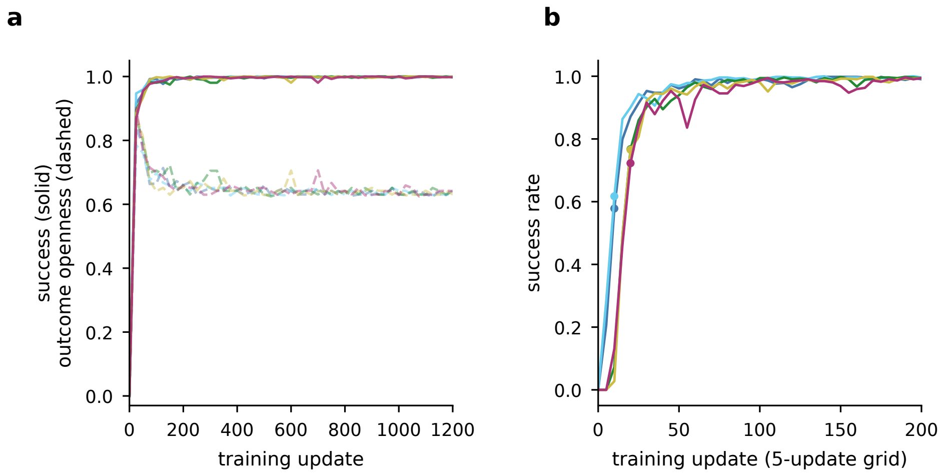


Figure 3. Capability formation is smooth across training. **a**, Formation axis (across training): success (solid) and outcome openness (dashed) rise smoothly, no breakpoint (0/5). **b**, At fine 5-update resolution the success rise is fast (per-seed midpoints, dots, at updates 10–20) but still has no structural onset (0/5). Formation is smooth and expansive while realization within an episode is punctuated (Fig. 2b)—same system, same instrument.

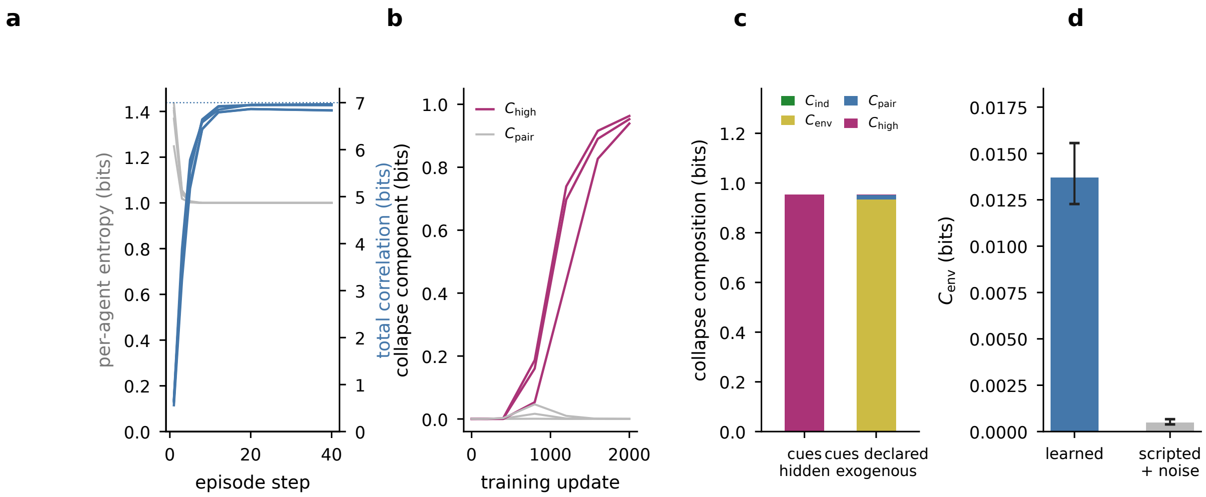


Figure 4. The source typology transfers to learning. **a**, Hidden-coordination task (five training seeds): per-agent entropy stays open while total correlation rises to near the 7-bit maximum (dotted line)—a purely relational collapse; lines show individual seeds. **b**, With the low-order route blocked, learning builds an irreducible higher-order (XOR) carrier: C_{high} grows while C_{pair} stays near zero (three training seeds; lines show individual seeds). **c**, Contract-relativity on the same three seeds: declaring the private cues exogenous re-attributes the identical collapse from C_{high} to C_{env}; bars show seed means. **d**, At matched task score, the environment-conditioned component separates a learned pair from a noise-matched scripted pair (30 episodes per layout; error bars are 95% episode-bootstrap confidence intervals from 1,000 resamples).

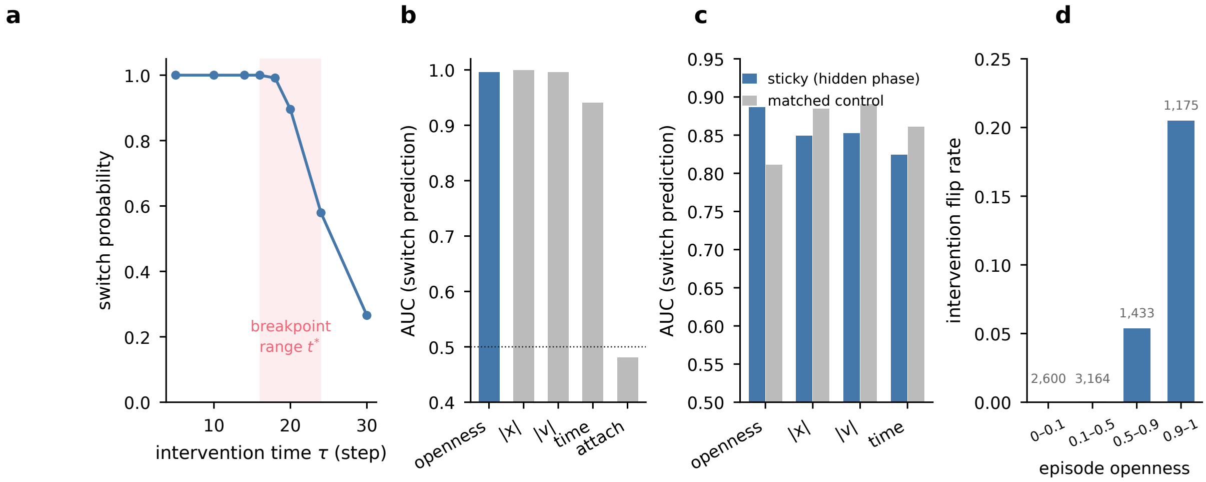


Figure 5. Remaining openness predicts controllability. **a**, Grip system: probability that a counter-regime impulse switches the outcome, as a function of intervention time; the window closes after the breakpoint range (shaded). Points pool 2,048 evaluation episodes per seed and time across five trained seeds. **b**, Openness predicts switchability as well as or better than physical baselines; bars show pooled AUC across the same seeds and intervention times. **c**, Matched-parameter causal contrast: openness beats the order parameter only when a hidden consolidation phase exists (sticky), and the ranking reverses in the control; bars show pooled AUC across five training seeds per condition. **d**, Ant colony, paired counterfactuals: intervention flip-rate rises with the episode's own remaining openness; closed episodes never flip ($n = 8{,}372$ episode pairs; numbers above bars give bin counts).

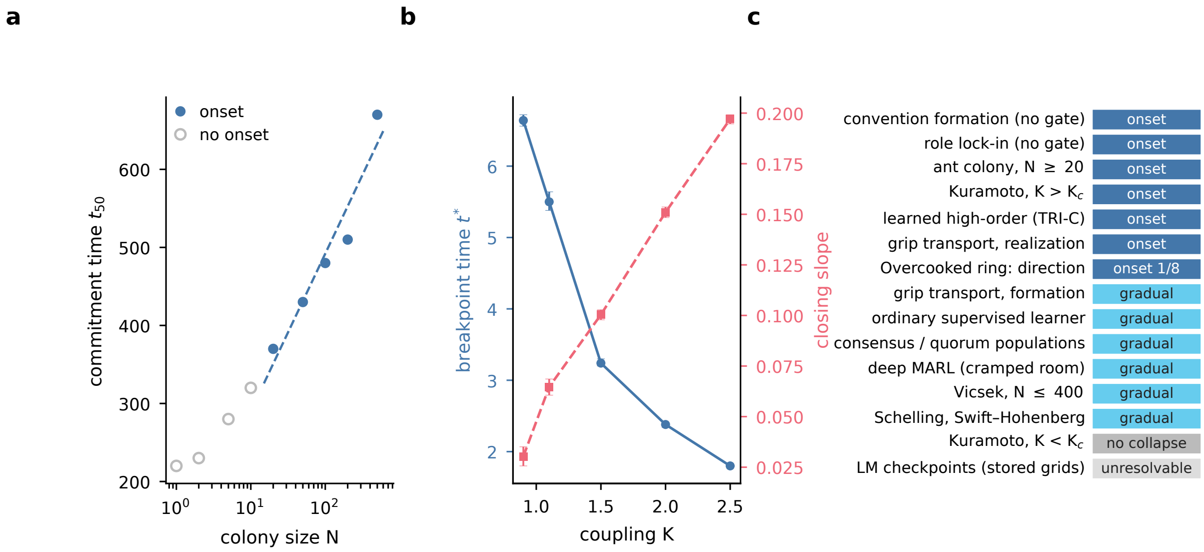
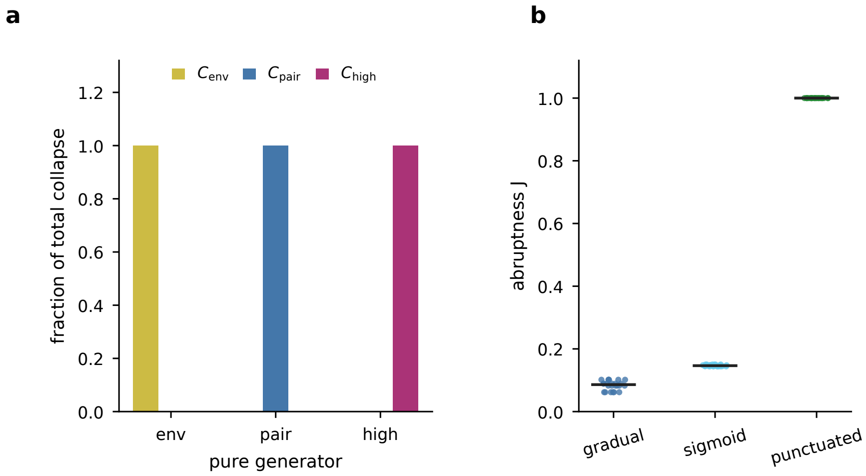
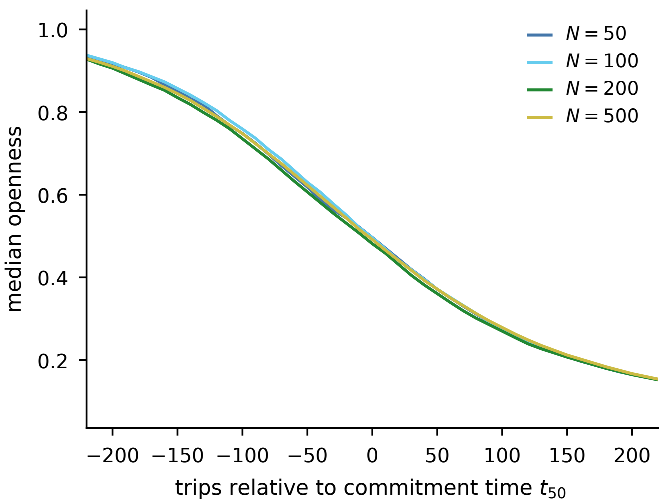


Figure 6. Scaling and scope. **a**, Finite-size scaling in the ant model: commitment time is consistent with $t_{50} = a + b \ln N$ ($b = 87.6$, 95% CI [41, 124], $R^2 = 0.93$; dashed fit), as predicted by nucleation from $N^{-1/2}$ fluctuations; open circles, sizes below the onset threshold. Each size uses 30 episodes; the slope interval is an episode-bootstrap 95% confidence interval from 200 resamples (see Extended Data Fig. 2 for the universal collapse profile). **b**, Kuramoto criticality: breakpoint time falls and closing slope rises as coupling increases; points are means of ten seeds per coupling, error bars are 95% bootstrap intervals over seeds, and subcritical runs are gated null. **c**, Classification: onset appears in systems that cross a joint-regime barrier, including convention formation and role lock-in without a designed gate. In the standard Overcooked coordination-ring layout, all 8 seeds commit to a circulation direction, but only 1/8 certifies monotone onset (Results). The detector reports gradual convergence for formation, ordinary learners, consensus/quorum populations, cramped-room deep MARL and several classic cases.



Extended Data Fig. 1. Instrument validation on constructed ground truth. a, Source-decomposition ladder on analytic ground truth: each pure generator loads almost entirely onto its own component. **b**, Full-factorial calibration (72 cells): amplitude M is invariant across temporal shape while abruptness J strictly orders punctuated $>$ sigmoid $>$ gradual.



Extended Data Fig. 2. Finite-size universality of the collapse profile. Median openness trajectories for ant colonies of $N = 50$–500, aligned by commitment time t_{50}: the four profiles superpose under pure time translation (mean pairwise RMS 0.010 versus 0.266 unaligned). Increasing N shifts the timing but does not reshape the collapse.